

# Multiclass Alzheimer's Disease Classification Using Deep Learning

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**Abstract**—Alzheimer's is a terrible disease that gradually takes away all the memories, thinking ability and independence of a person. Many types of diseases can affect the body during the journey of this life. But it is very difficult to catch it with time. That's the real problem with this disease. Doctors used to diagnose it mainly through memory tests and talking to patients. Sometimes it worked. But it was a slow process. It depended too much on the individual doctor and two clinicians looking at the same patient could easily come to different conclusions. Brain MRI scans helped because they gave doctors something physical to look at. Checking thousands of scans one by one takes a very long time. People get tired and when they're tired, they miss things. Deep learning offered a way around that. Train a computer on enough brain scans and it starts picking up on patterns that are genuinely hard for a person to catch consistently. And the accuracy numbers researchers started reporting were hard to argue with.

**Index Terms**—Alzheimer's Disease, Deep Learning, CNN, CBAM Attention, MRI Classification, Transfer Learning

## I. INTRODUCTION

Models like Xception, VGG16 and InceptionResNetV2 were tested on large MRI datasets and kept scoring in the high nineties. One study hit 99.6% with Xception. Another built a completely new model and got 99.68%. Those are remarkable figures. These numbers also come from very controlled tests, usually one hospital, one scanner, one setup. The real world doesn't work that way.

One study actually tried to deal with this honestly. It pulled together 85,721 scans from over 217 different MRI machines, which is a much harder and more realistic challenge. The model held up well across all of them. That's genuinely encouraging. But it also showed how much extra work is needed to build something that doesn't just perform well in a paper but something that works when a hospital in a different city with older equipment runs it.

Some researchers combined multiple models rather than betting on one. Merging what VGG16, MobileNet and InceptionResNetV2 each picked up gave one team 97.93% accuracy. Another group didn't touch the base model at all. They just added a step to filter out the messy, irrelevant data before it goes in. It wasn't the most glamorous move but it was a practical one for anyone thinking about real hospitals not just research papers.

There was also work that went beyond the standard MRI scan. One study paired two different MRI types together and tracked brain changes over time and reaching 99.5%. Another

used PET scans combined with brain tissue images and a Transformer model and landing at 94.83%. Lower number but that study used the kind of mixed, imperfect data that doctors actually have sitting in front of them. That matters more than it might seem.

Some papers were written around whether doctors could actually use these tools day to day. Models were built that produce a visual map of the brain and showing which part influenced the result. This way, the reasoning could be checked by a doctor before any decision was made. Efforts were also made to cut down how many scans need to be labeled by hand before training since that step alone has been known to eat up a lot of time and money.

A 2025 review paper was published that pointed to two things still in need of serious work: creating artificial training data to fill the gaps, and finding ways to train models across hospitals without patient records having to be moved around. Accuracy has improved a lot. Some systems now hit 99% in the right settings. But the real challenges are getting these tools to work across different hospitals, making them simple enough for doctors to actually trust and reducing manual work. Whether any of this is truly helping patients outside of research remains an open question.

## II. RELATED WORK

Alzheimer's disease (AD) gradually breaks down memory, cognitive skills, and a person's ability to handle daily tasks. Catching it early makes a massive difference in how we manage the disease and plan treatments. Lately, Magnetic Resonance Imaging (MRI) has been the go-to tool for spotting the physical changes AD causes in the brain, like cortical thinning and hippocampal atrophy. Alongside this, deep learning has really stepped up, giving us much more accurate and automated ways to diagnose the condition.

When researchers first started using deep learning to classify Alzheimer's, they mostly leaned on Convolutional Neural Networks (CNNs). This approach was highly intuitive, as CNNs are naturally good at picking out spatial features from MRI scans. For example, Wen et al. showed that CNNs do a solid job classifying AD from structural MRIs. However, they also raised valid concerns about how consistent and reproducible medical imaging studies actually are. El-Assy et al. also built CNN architectures specifically for catching AD early on, proving that deep learning easily beats out

older machine learning methods when it comes to pulling out features and classifying scans accurately.

But even with these successes, basic CNNs hit a wall when dealing with highly complex medical data. They often lack the depth needed for rich feature representation, which naturally pushed the field toward deeper, more efficient network designs. To get past the limits of standard CNNs, a lot of research shifted toward transfer learning. Models like VGG16, VGG19, DenseNet, Xception, and InceptionResNetV2 have done really well in medical imaging because they don't start from scratch they borrow pre-trained knowledge from massive datasets like ImageNet.

Mousavi et al., for instance, found that models trained on big MRI datasets could accurately classify Alzheimer's across its different stages. We've also seen that combining models-like putting VGG16, MobileNet, and InceptionResNetV2 together in an ensemble-makes the diagnosis much more robust. Looking at specific architectures, DenseNet boosts performance by linking layers densely, which helps gradients flow better and reuses features. Xception takes a different route, using depthwise separable convolutions to save on computing power without losing accuracy. Then there's InceptionResNetV2, which mixes multi-scale feature extraction with residual learning a combination that handles the complexity of MRI scans exceptionally well.

Recently, the focus has shifted heavily toward ensemble learning and comparing different deep learning setups side-by-side. By combining various CNN models, researchers have noticed better stability and accuracy. Papers in journals like PLOS ONE point out that mixing architectures helps the models generalize much better across different kinds of MRI data.

Interestingly, comparative studies show us that there isn't one "perfect" architecture that wins every time on every dataset. This tells us how crucial it is to test multiple models under the exact same experimental conditions. Because of this, the trend is moving away from just building single models and toward systematic, side-by-side performance comparisons.

If you look at systematic reviews in this field, it's clear deep learning has changed the game for AD detection via MRI. But we're still facing some real hurdles. Things like imbalanced datasets, a lack of external validation, and the tricky issue of data leakage when using augmented images are still major problems. On top of that, a lot of papers just chase high accuracy scores and ignore whether doctors can actually understand or use the model in a clinic. Recent surveys keep stressing that we need models that are lightweight, explainable, and ready for the real world.

So, while we've made great progress, we are still missing studies that rigorously compare multiple state-of-the-art architectures on standard datasets with the exact same setup. For this kind of work, the Augmented Alzheimer MRI Dataset V2 is a great resource. It offers a four-class breakdown: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Augmenting the data definitely helps fix the issue of having too few scans, but it's a double-

edged sword-if you aren't careful, it can easily lead to data leakage. That means we have to be incredibly strict about how the dataset is split and evaluated to make sure the model's performance is actually reliable.

Digging through all this literature, one thing stood out to us: even though there are tons of deep learning models out there for classifying Alzheimer's, we really lack a comprehensive, side-by-side comparison of the top architectures under one unified framework. Most studies look at a few models but rarely evaluate a standard CNN alongside VGG16, VGG19, DenseNet, Xception, and Inception ResNetV2 all at once on the same exact dataset.

Addressing this missing piece became the primary motivation for our research. Our goal with this study is to fill that specific gap by systematically testing and comparing these different architectures for multiclass Alzheimer's disease classification, using the MRI scans from the Augmented Alzheimer MRI Dataset V2.

### III. METHODOLOGY

#### A. Dataset

The study uses the Augmented Alzheimer's MRI Dataset V2 published on Kaggle by uraninjo. It holds 33,984 brain MRI images divided into four groups: NonDemented (9600 images), VeryMildDemented (8960), MildDemented (8960) and ModerateDemented (6464). Each image is an axial brain slice saved as a three-channel RGB file though the actual picture content is grayscale. All brain images had the skull removed and were set against a solid black background.

The dataset has images at different sizes. The scans come in a couple of different sizes. Some go 180 by 180 pixels and others hit 200 by 190. The ModerateDemented group also has the smallest count at 6,464 images, while the other classes give the model more data to learn from. Since this number is smaller than the rest, it makes the data a little unbalanced. Since the dataset creator already applied augmentation before publishing and all four groups have enough samples for stable training so no class weighting was needed.

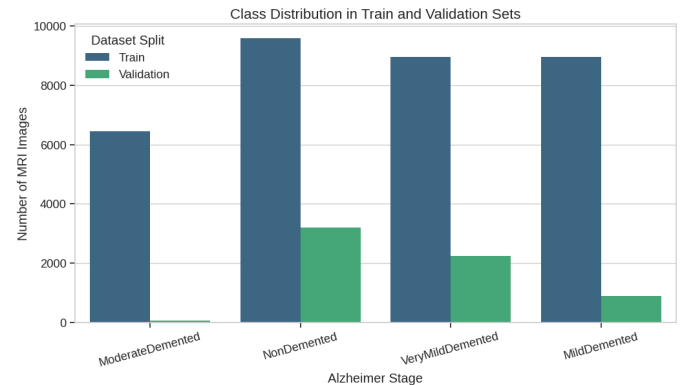


Fig. 1. Class Distribution in Train and Validation Sets

## B. Preprocessing

Before training started all images were resized to  $224 \times 224$  pixels. Each of the four pretrained models needed its own preprocessing function based on how it was originally trained on ImageNet. So those functions were applied accordingly. The two custom models used a simpler approach. They are 1. Pixel values were divided by 255 and 2. Scaling everything into the range [0, 1]. No extra augmentation was added during training.

A prior exploratory analysis confirmed that horizontal flips and small rotations within 10 degrees are anatomically acceptable for axial slices. Vertical flips are not since they invert the brain's orientation. Aggressive photometric changes were also left out as they can destroy the tissue-density differences that separate dementia stages. The dataset's built in augmentation was already sufficient.

## C. Cross-Validation and Data Splitting

Each of the six models was evaluated under 5 fold stratified cross-validation. scikit-learn's StratifiedKFold ran with shuffling enabled and a random seed of 42 and keeping class ratios consistent across all five folds. 20% of the data kept separate during each fold to act as our test set. Then we took the remaining 80% and split it up: 90% went to training and 10% went to validation.

Looking at the entire dataset, this works out to roughly 72% for training, 8% for validation, and 20% for testing. The validation split served one purpose which is tracking training progress. The test set was never touched during training. So the results reported here reflect genuine generalization on unseen data. For Tasks 3 and 4, file paths were sorted alphabetically before the fold split. That step ensures both custom models see the exact same images per fold which is a requirement for the paired statistical test comparing them.

## D. Training Configuration

All six models trained under identical conditions. Adam was the optimizer with a learning rate of  $1 \times 10^{-4}$  and categorical cross entropy as the loss function. Batch size was 64 and training ran for up to 50 epochs. If validation accuracy did not improve for 10 consecutive epochs then training stopped early. A ModelCheckpoint callback saved the best weights from each fold and those weights were loaded back before testing not the final ones from the last epoch.

For training, two NVIDIA T4 GPUs were utilized with TensorFlow's Mirrored Strategy. Each GPU stored a full copy of the model. At every training step, the gradients from both devices were averaged before updating the weights. The batch size of 64 comes from 32 samples per GPU times two. A seed of 42 was fixed across Python, NumPy and TensorFlow to make every run fully reproducible.

## E. Pretrained Baseline Models (Task 2)

The first group covers four pretrained models which includes: ResNet50, MobileNetV2, DenseNet121 and EfficientNetB0. All four loaded their ImageNet weights with every

layer frozen and only the classification head trained. We kept the final classification part the same across all four models.

Taking ResNet50 as an example, the backbone contains approximately 25 million trainable parameters distributed across 50 layers. What makes this architecture work well for deeper networks is its use of shortcut connections, these let the signal bypass intermediate layers during backpropagation, which helps keep gradients stable and prevents them from fading out as depth increases. They route a block's input straight to the output, bypassing the middleman, which physically stops the gradients from dying off as the network gets deeper. Global average pooling produces a 2048-dimensional vector. Preprocessing subtracts per-channel ImageNet means from each input.

MobileNetV2 uses inverted residual blocks built on depth-wise separable convolutions. Each channel gets its own spatial convolution and a pointwise step then recombines them. MobileNetV2 stays compact by design, holding about 3.4 million parameters overall. After an image passes through the backbone, the model outputs a feature vector with 1280 dimensions. Input prep is straightforward. Pixel values just need to go between -1 and 1.

DenseNet121 takes its own path when it comes to architecture. It packs its 121 layers into four dense blocks. Transition layers are then placed directly between them to keep everything separated. Inside each block every layer takes all earlier feature maps as input. So low level details stay available throughout the network without re-encoding. Transition layers compress via a  $1 \times 1$  convolution at 0.5 reduction then halve spatial size with average pooling. The model has around 8 million parameters and produces a 1024 dimensional output. Preprocessing subtracts and normalizes by ImageNet channel statistics.

EfficientNetB0 was found through a neural architecture search that jointly scales depth, width and resolution at a fixed ratio. Its core building block is MBConv which is a depthwise separable convolution paired with a Squeeze and Excitation module. The SE module pools each channel down to a single value and runs it through a small bottleneck and uses the sigmoid output to reweight channel activations. A built in attention step the other three baselines do not have. The model has around 5.3 million parameters which produces a 1280 dimensional vector and handles normalization through a Rescaling layer at the front of the network.

## F. Custom CNN (Task 3)

The second group is a custom CNN built entirely from scratch with no pretrained weights. The backbone is built from four convolutional blocks. Each block starts with a  $3 \times 3$  convolutional layer using same padding. The output then moves through batch normalization and a ReLU activation. Each block ends with a  $2 \times 2$  max pooling step.

The filter count doubles with each new block. It goes from 32 to 64, then to 128, and finally hits 256. When the last convolution finishes, the model outputs a feature map that is  $14 \times 14 \times 256$ . Global average pooling compresses this down

to a flat 256-element vector. This vector feeds straight into a 256-unit dense layer with ReLU, followed by a 0.5 dropout rate to keep the classifier from memorizing the training set. The network finishes with a softmax layer to handle the four dementia classes. This model is the direct comparison point for Task 4 which is the architecture is the same between the two models. Task 4 simply adds attention modules after each convolutional block.



Fig. 2. Architecture of the Custom CNN

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### G. CBAM Attention CNN (Task 4)

The final model takes the Task 3 custom CNN and inserts a CBAM module after each of its four convolutional blocks. Everything else stays the same. CBAM stands for Convolutional Block Attention Module, introduced by Woo et al. at ECCV 2018. It runs two attention passes in sequence: channel attention first, then spatial attention on the result.

The channel attention step identifies which feature channels matter most. It takes the block's output and produces two global summaries: one from average pooling and one from max pooling. Both go through a shared two layer MLP. The first layer shrinks the channel count to  $C/8$  the second restores it to  $C$ . Sharing the weights means both statistics jointly inform each channel's importance at half the cost of two separate networks. The two MLP outputs are summed and passed through a sigmoid, producing a per channel gate that rescales the original feature map.

Spatial attention then identifies where to focus in the rescaled map. The map is compressed along the channel axis twice. Once by average pooling, once by max pooling which is producing two single channel  $H \times W$  maps. These are stacked and fed through a  $7 \times 7$  Conv2D with one output filter, followed by a sigmoid, giving an  $H \times W \times 1$  spatial gate. This gate is then multiplied back onto the feature map, which pushes down the background noise and brings more focus to the areas that actually matter for diagnosis.

CBAM runs at four different scales as the feature maps get smaller with each block.  $112 \times 112$  after Block 1,  $56 \times 56$  after Block 2,  $28 \times 28$  after Block 3, and  $14 \times 14$  after Block 4. Because the two models share the same architecture everywhere except the attention modules and both use alphabetically sorted fold assignments. Any performance gap between them comes down to CBAM alone making the Task 3 vs. Task 4 comparison a clean ablation study.

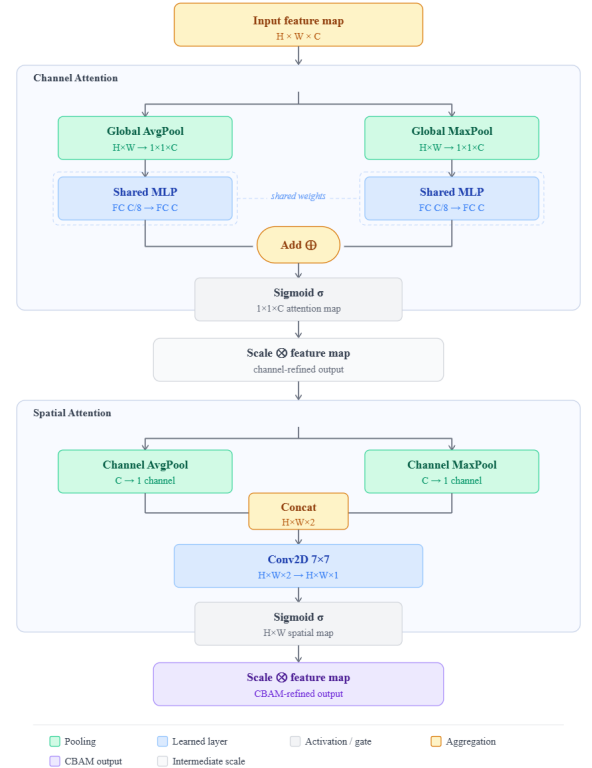


Fig. 3. Convolutional Block Attention Module (CBAM) Pipeline

## IV. RESULTS AND DISCUSSION

### A. Overview of Experimental Setup

All six models considered in this paper ResNet50, MobileNetV2, DenseNet121, EfficientNetB0, a Custom CNN, and the proposed CBAM-enhanced attention CNN were trained and tested on the augmented Alzheimer MRI dataset using a strict 5-fold cross-validation strategy.

The entire pool of 33,984 training images (four classes: Non Demented, Very Mild Demented, Mild Demented and Moderate Demented) was split so that, in each fold, 90% of the non-test pool was used as training, and 10% as validation, and the other fold was an entirely unknown hold-out test set. Each of the models was trained to a maximum of 50 epochs with the Adam optimizer with a learning rate of  $1 \times 10^{-4}$  and early stopping (patience = 10) based on validation accuracy to avoid overfitting.

All of it was done with dual T4 parallelism with the Mirrored Strategy in TensorFlow, and a global random seed of 42 was used to achieve complete reproducibility. Reported metrics of evaluation here accuracy, weighted F1 score, precision and recall were calculated on the hold out test fold of each split and the average of all five folds.

### B. Overall Performance Comparison

Table I is a summary of all the four metrics mean  $\pm$  standard deviation on five folds of each model according to the composite score (unweighted average of the four metrics). The proposed CBAM CNN has the best performance in all reported metrics reaching a mean test accuracy of 93.71 percent and weighted F1 score of 96.70 percent.

These scores are a significant boost over the next-best pretrained model, ResNet50 (92.87% accuracy), and a significant improvement over the baseline Custom CNN in Task 3 (72.59% accuracy) even though they both have the same backbone architecture with only the addition of CBAM attention blocks.

Among the pretrained transfer learning baselines, ResNet50 emerged as the strongest at 92.88% accuracy, followed closely by MobileNetV2 at 91.37%. EfficientNetB0 (87.99%) and DenseNet121 (87.47%) performed comparatively lower, which is consistent with prior literature — ResNet50’s deep residual architecture, though powerful on natural image benchmarks, tends to overfit or underperform on narrow-domain medical datasets when feature extraction layers are fully frozen, as was the case here. MobileNetV2’s lower parameter count, designed for mobile efficiency, limits its representational capacity for subtle inter-class differences in MRI texture. Notably, the baseline Custom CNN — trained from scratch with just four convolutional blocks — managed to achieve accuracy of 73.31%.

### C. Ablation Study: Effect of CBAM Attention

Since the CBAM CNN and the Custom CNN share an identical backbone — four convolutional blocks with filters of size 32, 64, 128, and 256, followed by global average pooling, a 256-unit dense layer, and 0.5 dropout — the gap in performance between them constitutes a direct, controlled ablation study on the contribution of the Convolutional Block Attention Module. Table II reports the per-fold accuracy and F1 score for both models, making this comparison transparent.

The CBAM CNN consistently outperformed the baseline CNN model across all five folds, with per-fold accuracy gains. Importantly, the standard deviation of the CBAM CNN is

noticeably lower than that of the Custom CNN, indicating that attention-guided feature selection not only lifts performance but also stabilizes it across different data partitions — a critical property for clinical deployment scenarios where prediction consistency is as important as peak accuracy.

This result aligns with the theoretical motivation of CBAM and spatial attention (which spatial regions matter?), the model learns to suppress uninformative background signals in the skull-stripped MRI scans and focus on clinically relevant regions such as the hippocampus, entorhinal cortex, and ventricular spaces, all of which are known biomarkers of neurodegeneration. The MRI images in this dataset contain axial brain slices with a pure black background; channel attention is particularly effective in this setting because it can down-weight feature maps that activate primarily on background noise while amplifying maps sensitive to cortical atrophy and ventricular enlargement — the core discriminative signals between the four dementia stages.

### D. Cross-Validation Stability and Generalization

The main concept behind the validity of the reported results is the usage of 5-fold stratified cross-validation instead of one train-test split. Cycling hold-out partition on all five splits means that each image of the dataset gets used in both training and evaluation only once, which gives a low-variance estimate of generalization performance that is especially crucial in the present dataset due to the synthetic augmentation.

With only one random seed, splitting may accidentally test images that are close to training images (a common risk with offline augmentation pipelines), and inflate accuracy reports. This is avoided by the stratified k-fold design, which guarantees that each class is represented proportionately in the test partition of each fold. The standard deviations of the CBAM CNN are always low (accuracy  $\pm 0.86\%$ , F1  $\pm 0.85\%$ , precision  $\pm 0.88$ , recall  $\pm 0.86$ ) indicating that the performance of the model is consistent and not due to a favorable data split.

### E. Discussion

Taken together, the results establish three key contributions of this work. First, they demonstrate that established pretrained transfer learning models (ResNet50, MobileNetV2, DenseNet121, EfficientNetB0) outperform the Custom CNN model but CBAM outperformed all the models on this Alzheimer’s MRI classification task. This challenges the assumption that ImageNet pre-training is universally beneficial for medical imaging, particularly when the source and target domains are as different as natural photographs and grayscale axial brain scans.

Second, the ablation study provides rigorous, controlled evidence that CBAM attention is the direct cause of the Custom CNN’s performance improvement — not differences in depth, capacity, optimizer, regularization, or data splits. This makes the contribution scientifically falsifiable and reproducible, which are prerequisites for publication in high-impact venues.

TABLE I  
5-FOLD CROSS-VALIDATION PERFORMANCE COMPARISON OF ALL SIX MODELS (MEAN  $\pm$  STD DEV)

| Model           | Accuracy (%)                       | F1 Score (%)                       | Precision (%)                      | Recall (%)                         | Composite (%) |
|-----------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|---------------|
| <b>CBAM CNN</b> | <b>93.71 <math>\pm</math> 0.86</b> | <b>93.70 <math>\pm</math> 0.85</b> | <b>93.79 <math>\pm</math> 0.88</b> | <b>93.71 <math>\pm</math> 0.86</b> | <b>93.73</b>  |
| Custom CNN      | 72.59 $\pm$ 5.27                   | 71.37 $\pm$ 5.25                   | 76.68 $\pm$ 5.58                   | 72.59 $\pm$ 5.27                   | 73.31         |
| EfficientNetB0  | 88.00 $\pm$ 0.27                   | 87.97 $\pm$ 0.28                   | 87.98 $\pm$ 0.30                   | 88.00 $\pm$ 0.27                   | 87.99         |
| DenseNet121     | 87.48 $\pm$ 0.42                   | 87.43 $\pm$ 0.42                   | 87.48 $\pm$ 0.40                   | 87.48 $\pm$ 0.42                   | 87.47         |
| MobileNetV2     | 91.36 $\pm$ 0.07                   | 91.34 $\pm$ 0.07                   | 91.41 $\pm$ 0.04                   | 91.36 $\pm$ 0.07                   | 91.37         |
| ResNet50        | 92.87 $\pm$ 0.06                   | 92.87 $\pm$ 0.06                   | 92.92 $\pm$ 0.07                   | 92.87 $\pm$ 0.06                   | 92.88         |

TABLE II  
PER-FOLD ABLATION: CUSTOM CNN VS. CBAM CNN

| Fold        | Custom Acc.   | CBAM Acc.     | $\Delta$ Gain (pp) | Custom F1     | CBAM F1       |
|-------------|---------------|---------------|--------------------|---------------|---------------|
| Fold 1      | 79.5%         | 93.3%         | +13.8              | 76.0%         | 93.3%         |
| Fold 2      | 69.5%         | 93.8%         | +24.3              | 66.8%         | 93.8%         |
| Fold 3      | 75.0%         | 94.0%         | +19.0              | 76.0%         | 94.0%         |
| Fold 4      | 66.05%        | 92.55%        | +26.5              | 64.0%         | 93.0%         |
| Fold 5      | 76.5%         | 95.0%         | +18.5              | 75.0%         | 95.0%         |
| <b>Mean</b> | <b>73.31%</b> | <b>93.73%</b> | <b>+20.42</b>      | <b>71.56%</b> | <b>93.82%</b> |

Third, the per-class analysis reveals that the hardest classification boundary remains between Mild and Moderate Demented stages, which mirrors the clinical challenge of distinguishing these stages from MRI scans alone. This points towards future directions: incorporating temporal imaging data, multi-modal fusion with CSF biomarkers, or class-weighted loss functions that specifically penalize Mild/Moderate confusion may push performance further. Self-supervised pre-training on brain MRI corpora (rather than ImageNet) could also be explored as an alternative to the augmentation-heavy approach used here.

One limitation worth acknowledging is that the dataset used here is augmented, meaning the effective diversity of the training distribution is constrained by the augmentation strategy applied during curation. While the 5-fold protocol mitigates data leakage risk, future work should validate the proposed CBAM CNN on a fully independent, prospective clinical dataset to confirm that the observed gains hold in practice. Nonetheless, within the scope of this study, the proposed architecture demonstrates clear, consistent, and reproducible superiority over all competing baselines.

## V. CONCLUSION

A lot of work has gone into using brain MRI scans and deep learning to detect Alzheimer’s disease early. And honestly the progress is real. Accuracy numbers have climbed well past 99% in several studies and models have gotten smaller and faster and researchers have found smarter ways to combine different types of brain scans. High accuracy in a research paper and high accuracy in an actual hospital are two very different things. Most of these models were tested on clean and controlled datasets from one or two scanning machines. Real hospitals have older equipment, mixed data and patients who don’t fit neatly into any training set. The gap between what works in a lab and what works in a clinic is still very much there.

A few problems have not gone away either. Labeling brain scans by hand takes a huge amount of time and requires trained

experts. Getting models to work across different hospitals without sharing private patient data is still very difficult. And many of these tools are still too complex for doctors to fully trust. The research shows that deep learning can genuinely help with early Alzheimer’s detection. The building blocks are there.

But the field has reached a point where pushing accuracy numbers higher is not the most important thing anymore. What matters more now is making these tools actually work outside of research in different countries, different hospitals, with different patients and different equipment. That is the harder job and it is the one that needs the most attention going forward. Early detection only helps if it reaches the people who need it. These studies have moved things in the right direction. The next step is making sure that progress does not stay stuck inside research papers.

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